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2027 CYCLE

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CUET UG

Mathematics

Differential Equations

Order, Degree, Separable, Homogeneous and Linear · Section A 4 + B1 III · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Order and degree	"Order and degree of differential equations." (Section A Unit 4, compulsory) — "Definition, order and degree, general and particular solutions of a differential equation." (Section B1 Mathematics Unit III)	Reading order and degree from raw equations, including the not-defined cases.
Variable separable equations	"Solution of differential equations by method of separation of variables." (Section B1 Mathematics Unit III; Section A Unit 4 lists "Solving of differential equations with variable separable")	Splitting variables cleanly and integrating both sides without losing $+C$.
Homogeneous equations	"Solutions of homogeneous differential equations of first order and first degree." (Section B1 Mathematics Unit III)	Spotting the $\frac{y}{x}$ ratio shape; running the $y = vx$ substitution end to end.
Linear equations and the integrating factor	"Solutions of linear differential equation of the type: $\mathrm{d}\frac{y}{\mathrm{d}}x + Py = Q$, where P and Q are functions of x or constants. $\mathrm{d}\frac{x}{\mathrm{d}}y + Px = Q$, where P and Q are functions of y or constants." (Section B1 Mathematics Unit III)	Building $e^{\int P}$ and finishing $y \cdot \mathrm{IF} = \int Q \cdot \mathrm{IF}$ in two lines.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus places this unit as **Section A Unit 4 (compulsory for every candidate)** and inside **Section B1 Unit III** — a compact family of exactly three solving methods, which makes it one of the highest technique-per-minute returns in the paper once the ladder below is memorised.

What a Differential Equation Is — and How to Measure It

A differential equation ties a function to its own derivatives: $\mathrm{d}\frac{y}{\mathrm{d}}x = 2x$ is the world's simplest example, and every growth, decay and cooling law in science wears this shape. Two measurements classify any equation. The **order** is the highest derivative that appears. The **degree** is the power of that highest derivative — with one strict precondition: the equation must be **polynomial in**

its derivatives. If a derivative sits inside a square root, a fraction power, or a sine, the degree is simply **not defined** — order still exists, degree does not. MCQs test exactly this boundary.

READING ORDER AND DEGREE — THREE GRADED EXAMPLES

$$\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^5 = 0 \quad \rightarrow \quad \text{order } 2, \text{degree } 3$$

$$\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{\frac{3}{2}} = 0 \quad \rightarrow \quad \text{order } 1, \text{degree not defined (fractional power of a derivative)}$$

$$\left(\frac{d^2y}{dx^2}\right)^2 + y = 0 \quad \rightarrow \quad \text{order } 2, \text{degree } 2$$

- **General vs particular.** A **general solution** carries arbitrary constants (A, B, C) — it names a whole family of curves. A **particular solution** fixes those constants from given conditions and names exactly one curve. First-order equations have one arbitrary constant; second-order have two.
- **Verification is a two-step reflex.** Differentiate the candidate solution, substitute into the equation, and demand an identity. This reflex answers "which of the following is a solution?" questions without solving anything.
- **Constants count order.** One integration brings one $+C$; the general solution of a first-order equation holds exactly one arbitrary constant — another MCQ evergreen.

WORKED EXAMPLE — VERIFY WITHOUT SOLVING

Show that $y = Ae^x + Be^{-x}$ satisfies $y'' - y = 0$ for every choice of A and B.

Solution. Differentiate twice: $y' = Ae^x - Be^{-x}$, then $y'' = Ae^x + Be^{-x} = y$. Substitute: $y'' - y = y - y = 0$ — an identity. Notice the two constants A and B match the second order: two integrations, two constants. If an option claims $y = Ae^x$ alone is the **general** solution, it is wrong for exactly that reason — one constant cannot cover a second-order family.

Rung 1 — Variable Separable

The kindest equations arrive **pre-separated**: $\mathrm{d}\frac{y}{\mathrm{d}}x = f(x) \cdot g(y)$. Push every y-factor to the $\mathrm{d}y$ side and every x-factor to the $\mathrm{d}x$ side, integrate both sides independently, and carry one $+C$ on the x-side (where it conventionally lives). The habit that pays: **integrate in symbols first, simplify the constant afterwards** — expressions like $\ln|y| = \frac{1}{2}\ln(1+x^2) + C$ become clean curve families $y = C\sqrt{1+x^2}$ with one renaming step.

WORKED EXAMPLE — TWO SEPARABLES, START TO FINISH

Part 1: solve $\mathrm{d}\frac{y}{\mathrm{d}}x = \frac{xy}{1+x^2}$.

Solution. Separate: $\frac{dy}{y} = \frac{x\,dx}{1+x^2}$. Integrate: $\ln|y| = \frac{1}{2}\ln(1+x^2) + C$. Rename $C \to \ln C_1$ and combine logs:

$$y^2 = C(1+x^2) \quad \text{or} \quad y = C\sqrt{1+x^2}.$$

Check by differentiating: $2yy' = 2Cx$, so $y' = C\frac{x}{y} = x\frac{y}{1+x^2}$ — identity.

Part 2: solve $\mathrm{d}\frac{y}{\mathrm{d}}x = e^{x+y}$.

Solution. Split the exponent: $e^{x+y} = e^x \cdot e^y$. Separate: $e^{-y}\,dy = e^x\,dx$. Integrate: $-e^{-y} = e^x + C$, or in the tidy form

$$e^{-y} + e^x = C.$$

The moral: "not obviously separable" often just means "split the exponent first".

Rung 2 — Homogeneous Equations (the y/x Shape)

An equation is **homogeneous of first order and first degree** when it can be written $\mathrm{d}\frac{y}{\mathrm{d}}x = F\left(\frac{y}{x}\right)$ — everything a ratio of y to x. The quick test: replace $x \to \lambda x$, $y \to \lambda y$; if the expression does not flinch, it is homogeneous. The substitution $y = vx$ (so $\mathrm{d}\frac{y}{\mathrm{d}}x = v + x\frac{\mathrm{d}v}{\mathrm{d}x}$) converts the equation into a **separable** one in v and x — the ladder's rungs are designed to feed each other. Finish by integrating in v, then substituting $v = \frac{y}{x}$ back; answers left in v score nothing.

WORKED EXAMPLE — ONE SUBSTITUTION, THEN SEPARABLE

Solve $x\frac{dy}{dx} = x + y$.

Solution. Divide by x: $\mathrm{d}\frac{y}{\mathrm{d}}x = 1 + \frac{y}{x}$ — a pure $F\left(\frac{y}{x}\right)$ shape. Substitute $y = vx$: $v + x\frac{dv}{dx} = 1 + v$, so $x\frac{dv}{dx} = 1$. Separate and integrate:

$$\int \mathrm{d}v = \int \frac{dx}{x} \quad \to \quad v = \ln|x| + C.$$

Substitute back $v = \frac{y}{x}$:

$$y = x\ln|x| + Cx.$$

Three moves — divide, substitute, back-substitute — and the equation is done. This is the shape of nearly every homogeneous MCQ.

Rung 3 — Linear Equations and the Integrating Factor

A **linear** equation reads $\mathrm{d}\frac{y}{\mathrm{d}}x + Py = Q$ with P, Q functions of x (or constants) — y appears to the first power with no funny business. The engine is the **integrating factor** $\mathrm{IF} = e^{\int P\,\mathrm{d}x}$: multiply the whole equation by it and the left side becomes the derivative of $y \cdot \mathrm{IF}$ in one stroke, leaving a single integration:

THE LINEAR ENGINE

$$\text{IF} = e^{\int P\,dx} \quad \rightarrow \quad y \cdot \text{IF} = \int (Q \cdot \text{IF})\,dx + C$$

$$\text{Flipped form:} \quad \frac{dx}{dy} + Px = Q \quad \text{uses} \quad \text{IF} = e^{\int P\,dy}$$

WORKED EXAMPLE — THE ENGINE AT WORK, TWICE

Part 1: solve $\mathrm{d}\frac{y}{\mathrm{d}}x + \frac{y}{x} = x^2$.

Solution. $P = \frac{1}{x}$, so $\int P\,dx = \ln|x|$ and $\text{IF} = e^{\ln|x|} = x$. Multiply through: $xy' + y = x^3$, i.e. $(xy)' = x^3$. Integrate: $xy = \frac{x^4}{4} + C$, so

$$y = \frac{x^3}{4} + \frac{C}{x}.$$

Verify: $y' + \frac{y}{x} = 3\frac{x^2}{4} - \frac{C}{x^2} + \frac{x^2}{4} + \frac{C}{x^2} = x^2$ — identity.

Part 2: solve $\mathrm{d}\frac{y}{\mathrm{d}}x + 2y = e^{-x}$.

Solution. $P = 2$ (a constant is allowed), $\text{IF} = e^{2x}$. Then $\left(ye^{2x}\right)' = e^{2x} \cdot e^{-x} = e^{x}$, so $ye^{2x} = e^{x} + C$ and

$$y = e^{-x} + Ce^{-2x}.$$

Two integrations, two appearances of C handled in one stroke — the engine's whole point.

EVERYDAY EXAMPLE — THE BALANCED SEESAW

A linear equation is a seesaw: the term Py pushes y back toward balance (friction, leakage, cooling), while Q is the outside push (a pump, sunlight, an income). The integrating factor is the lever arm that makes the friction side an exact derivative — once you multiply by it, the left side **is** something you can integrate, no argument. If a solution ever looks almost right but the left side refuses to become a single derivative, the almost-certain cause is a missing or wrong IF — the lever was never attached.

Choosing the Rung in the Exam

Run the three questions in order. **Is it a product of an x-part and a y-part?** Rung 1, separate. **Is everything ratio-shaped and same-degree?** Rung 2, substitute $y = vx$. **Does y sit to the first power with a clean coefficient?** Rung 3, build the IF. The syllabus names exactly these three rungs — no differential equation in this unit can escape the ladder, which is why this chapter returns marks faster than almost any other once the shapes are memorised.

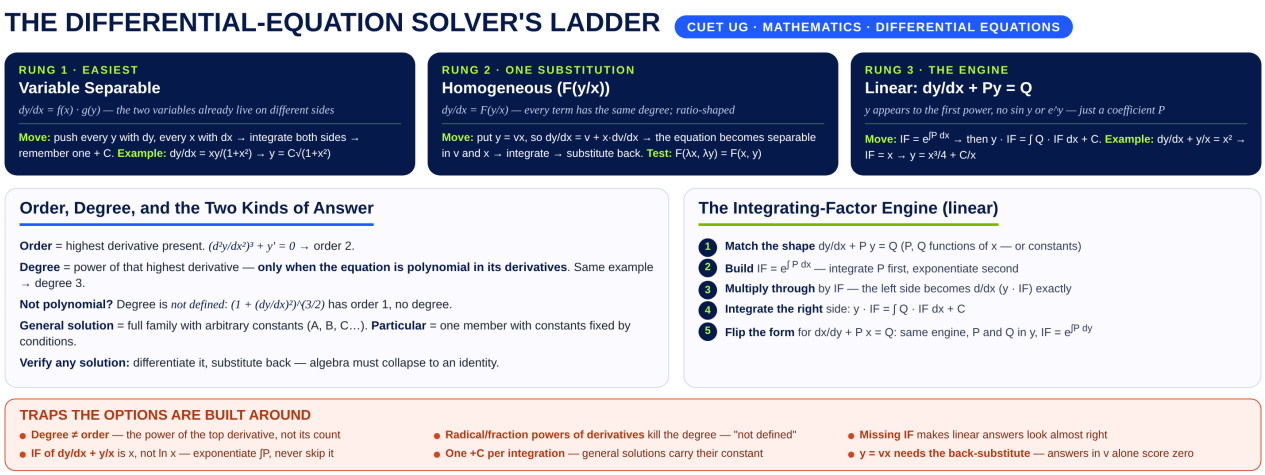


Figure 1: The solver's ladder: three rungs in difficulty order, the IF engine's five steps, and the traps waiting under each rung.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — DIFFERENTIAL EQUATIONS

Formula / rule	Letters mean	Fire it when
Order = highest derivative present	count the dashes	classification
Degree = power of top derivative, polynomial only	radicals kill it	classification MCQs
General solution: constants alive; particular: fixed	one C per order	solution-type questions
Verify: differentiate, substitute, expect identity	no solving needed	"which is a solution?"
Separable: $\frac{dy}{y} = f(x)\,dx$ form	split variables	product-shaped equations
$\int \frac{dy}{y} = \ln\lvert y\rvert$, $\int e^x = e^x$	standard forms	separable integrals
Homogeneous test: $F(\lambda x, \lambda y) = F(x, y)$	ratio shape	same-degree equations

Formula / rule	Letters mean	Fire it when
Homogeneous move: $y = vx$, $\mathrm{d}\frac{y}{\mathrm{d}}x = v + x\frac{dv}{dx}$	then separable	rung 2
Linear shape: $\frac{dy}{dx} + Py = Q$	P, Q in x or constants	rung 3 check
IF $= e^{\int P\,\mathrm{d}x}$	integrate P, exponentiate	linear engine
Solution: $y \cdot \mathrm{IF} = \int (Q \cdot \mathrm{IF})\,\mathrm{d}x + C$	one integration left	linear answers
Flipped: $\frac{dx}{dy} + Px = Q$, IF $= e^{\int P\,\mathrm{d}y}$	same engine in y	x-dependent forms

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — DIFFERENTIAL EQUATIONS

The mistake	What is right	The trap it sets
Reading degree as the count of derivatives	Degree is the **power** of the top derivative	Order/degree swapped options
Assigning degree to $\sqrt{d\frac{y}{d}x}$ forms	Not polynomial in derivatives — degree not defined	"degree = 1" bait
Skipping the $+C$ after one integration	Every integration births a constant	Options missing the constant family
Using $\int P$ itself as the IF	The IF is $e^{\int P}$ — exponentiate	The single most repeated IF trap
Linear check missed on $\sin y$ terms	Linear means y to power 1, no composites	Rung-3 applied to non-linear shapes
Answers left in v	Back-substitute $v = \frac{y}{x}$	Neat-looking wrong-variable options
Testing homogeneity by eye	Run $F(\lambda x, \lambda y) = F(x, y)$	Ratio-shaped non-homogeneous lures
Forgetting the flipped IF form	$\frac{dx}{dy} + Px = Q$ uses $e^{\int P\,\mathrm{d}y}$	Options with the x-engine applied blindly

Sixty-Second Self-Test

Statement	Verdict
The equation $y' + y^2 = x$ is linear.	False — y^2 breaks linearity
A first-order general solution carries two constants.	False — one integration, one constant
$\frac{dy}{dx} = e^{x+y}$ is separable after splitting the exponent.	True — $e^{x+y} = e^x e^y$
The IF of $y' + \frac{y}{x} = x^2$ is $\ln x$.	False — it is $e^{\ln x} = x$
Homogeneous answers may stay in terms of v.	False — substitute back $v = \frac{y}{x}$
$y = Ae^x + Be^{-x}$ solves $y'' - y = 0$.	True — differentiate twice, watch y return



Figure 2: The chapter in one look: classification, two solution kinds, three solving rungs — and the traps under each.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Identify the rung **before** computing — the choice of method is half the answer.

Q1. The order and degree of the equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^5 + y = 0$ are:

(A) order 3, degree 2 **(B)** order 2, degree 2
(C) order 2, degree 3 **(D)** order 3, degree 3

Q2. The degree of the differential equation $\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{\frac{3}{2}} = 0$ is:

(A) 1 **(B)** 3/2
(C) 2 **(D)** not defined

Q3. The general solution $y = Ae^x + Be^{-x}$ (A, B arbitrary) satisfies which differential equation?

(A) $y'' + y = 0$ **(B)** $y'' - y = 0$
(C) $y' - y = 0$ **(D)** $y'' - y' = 0$

Q4. The general solution of $\frac{dy}{dx} = \frac{xy}{1+x^2}$ is:

(A) $y = C(1 + x^2)$ **(B)** $y^2 = C(1 - x^2)$
(C) $y^2 = C(1 + x^2)$ **(D)** $y = Cx(1 + x^2)$

Q5. The general solution of $\frac{dy}{dx} = e^{x+y}$ is:

(A) $e^y + e^x = C$ **(B)** $e^{-y} - e^x = C$
(C) $e^{-y} + e^x = C$ **(D)** $e^y - e^x = C$

Q6. The general solution of $\frac{dy}{dx} + \frac{y}{x} = x^2$ is:

(A) $y = \frac{x^3}{4} + \frac{C}{x}$ **(B)** $y = \frac{x^3}{4} + Cx$
(C) $y = \frac{x^2}{4} + \frac{C}{x}$ **(D)** $y = x^3 + \frac{C}{x}$

Q7. The integrating factor of $\frac{dy}{dx} + 2y = e^{-x}$ is:

(A) $2x$ **(B)** e^{2x}
(C) 2 **(D)** e^x

Q8. The general solution of $x\frac{dy}{dx} = x + y$ is:

(A) $y = x\ln|x|$ **(B)** $y = \ln|x| + Cx$
(C) $y = x^2\ln|x| + C$ **(D)** $y = x\ln|x| + Cx$

Q9. The general solution of $\frac{dy}{dx} + y = 1$ is:

(A) $y = 1 + Ce^{x}$ **(B)** $y = 1 + Ce^{-x}$
(C) $y = Ce^{-x} - 1$ **(D)** $y = -1 + Ce^{x}$

Q10. The number of arbitrary constants in the general solution of a second-order differential equation is:

(A) 2 **(B)** 1
(C) 0 **(D)** depends on the equation

Detailed Solutions

Sol. Q1. Highest derivative: the second — order 2. Its power in the equation: 3 — degree 3. The lower-order term $(y')^5$ does not affect either reading. Order-location first, then power-reading: two questions in one.

Answer: (C)

Sol. Q2. The derivative sits inside a fractional power, $(\ldots)^{\frac{3}{2}}$ — the equation is not polynomial in its derivatives. Order is 1, but degree is **not defined** — the boundary case the syllabus names explicitly.

Answer: (D)

Sol. Q3. Differentiate twice: $y' = Ae^{x} - Be^{-x}$, $y'' = Ae^{x} + Be^{-x} = y$. So $y'' - y = 0$. Option (A) is the sin/cos signature; option (D) would need $y'' = y'$.

Answer: (B)

Sol. Q4. Separate: $\frac{dy}{y} = \frac{x\,dx}{1+x^2}$. Integrate: $\ln|y| = \frac{1}{2}\ln(1+x^2) + C$, so $y^2 = C(1+x^2)$. Options (A) and (B) break the algebra of the two logs.

Answer: (C)

Sol. Q5. Split: $e^{-y}\,dy = e^{x}\,dx$. Integrate: $-e^{-y} = e^{x} + C$, tidied as $e^{-y} + e^{x} = C$. The double-negative is where (B) and (A) try to catch you.

Answer: (C)

Sol. Q6. Linear rung: $P = \frac{1}{x}$, $\int P\,dx = \ln|x|$, IF $= x$. Then $(xy)' = x^3$, so $xy = \frac{x^4}{4} + C$ and $y = \frac{x^3}{4} + \frac{C}{x}$. Option (B) is the classic "constant rides on top instead of dividing" slip.

Answer: (A)

Sol. Q7. By definition: IF $= e^{\int P\,dx} = e^{\int 2\,dx} = e^{2x}$. Option (C) forgot to exponentiate entirely — the trap this chapter exists to defuse.

Answer: (B)

Sol. Q8. Rung 2: $y' = 1 + \frac{y}{x}$ is homogeneous. $y = vx$ gives $v + xv' = 1 + v$, so $xv' = 1$, $v = \ln|x| + C$. Back-substitute: $y = x\ln|x| + Cx$. Answers without the Cx term are particular, not general.

Answer: (D)

Sol. Q9. Linear with $P = 1$, IF $= e^x$. Then $(ye^x)' = e^x$, so $ye^x = e^x + C$ and $y = 1 + Ce^{-x}$. Quick check: as $x \to \infty$, $y \to 1$ — the steady state the seesaw settles to.

Answer: (B)

Sol. Q10. Each integration contributes one arbitrary constant; a second-order equation integrates twice. Two constants — matching options like $Ae^x + Be^{-x}$, which carries exactly A and B.

Answer: (A)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q1	Counted derivative occurrences instead of the highest	Circle the highest derivative first; everything else is scenery
(B) in Q2	Read the outer exponent 3/2 as a degree	Degree needs a polynomial in derivatives — check inside the radical
(A) in Q4	Dropped the log arithmetic on the y side	Write $\ln\|y\| = \frac{1}{2}\ln(\ldots)$ fully before exponentiating
(B) in Q6	Attached C as Cx	The IF division forces $\frac{C}{x}$ — track the division step
(C) in Q7	Stopped at $\int P = 2$	The IF is the exponentiated integral — always one more step
(A) in Q9	Growth sign flipped	Steady-state sanity: $y' + y = 1$ must relax to 1, not blow up

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